

Mathematical modelling of antimalarial drug resistance

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The effect of malaria on the developing world is devastating. Each year there are more than 200 million cases and over 400,000 deaths, with children under the age of five the most vulnerable. Ambitious malaria elimination targets have been set by the World Health Organization for 2030. These involve the elimination of the disease in at least 35 countries. However, these malaria elimination targets rest precariously on us being able to identify, diagnose and treat the disease appropriately.

In this talk, I will introduce several statistical and mathematical models for monitoring the emergence and spread of antimalarial drug resistance. Results will be presented from a geostatistical model that have generated spatio-temporal predictions of resistance based on prevalence data. These data are available only at discrete study locations and times. In this way, I will explain how the model output provides new insight into the spread of drug resistance and unveils new information that existing techniques cannot provide. I will then discuss how the results of these models have been used to update public health policy, showing how mathematics can help to inform strategies against malaria.